Amazon ML Challenge 2025: Smart Product Pricing Solution

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1. Executive Summary

We developed and ensembled three multi-modal models, each pairing a different **text transformer (DistilBERT, RoBERTa-Large, DeBERTa-Large)** with a **vision transformer (ViT, CLIP-Large)**. Our approach's innovation lies in using an optimized weighted average of these models' predictions. This ensemble strategy proved highly effective to produce a more accurate and generalized final prediction.

2. Methodology Overview

2.1 Problem Analysis

We interpreted this challenge as a multi-modal regression task, requiring the prediction of a continuous variable (**price**) from a combination of text, image, and structured numerical/categorical data.

Key Observations:

- **SMAPE**, which is the primary evaluation metric that measures relative (percentage) error, heavily penalizes errors on low-priced items. This informed our decision to predict the **logarithm of the price** (**log(price + 1)**), as minimizing standard error on a log-transformed target naturally optimizes for relative error on the original scale.

- **Structured `catalog_content`:** The `catalog_content` field is a structured record in string format. Using regular expressions, we parsed it into distinct, high-value features: **`Item Name`**, **`Bullet Points`, IPQ, Value** (e.g., 12.0), and **`Unit`** (e.g., Ounce).
- **Critical Importance of IPQ:** The text often contained an **Item Pack Quantity** (e.g., "Pack of 6", "12 Count"), which is a primary driver of price. We engineered a specific function to extract this into a dedicated numerical feature (`ipq`), treating **1** as the default if no pack size was mentioned.

2.2 Solution Strategy

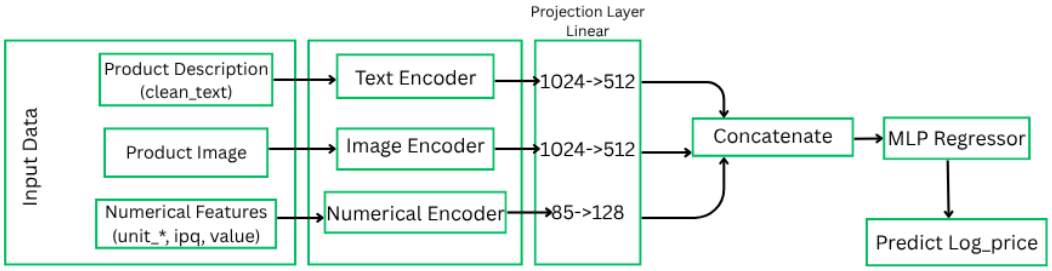
Our high-level approach is a **weighted ensemble of three distinct multi-modal deep learning models**. Each model is trained end-to-end, learning to simultaneously interpret features from text, images, and structured numerical data to predict the log-transformed price. The final submission is a weighted average of these three models' outputs. The

optimal weights were **determined empirically by minimizing the SMAPE score on a held-out validation set**, which produces a more robust and accurate final prediction than any single model achieved alone.

Approach Type: Weighted Ensemble of Multi-Modal Models
Core Innovation: Empirically optimizing the weighted average of three diverse, multi-modal transformer models (text and vision) to minimize validation SMAPE.

3. Model Architecture

3.1 Architecture Overview



3.2 Model Components

Text Processing Pipeline:

- **Preprocessing steps:**
  - The raw `catalog_content` string was **parsed using regular expressions** to extract structured fields: **`item_name`, `bullet_points`, `value`, `unit`, and `Item Pack Quantity (IPQ)`.**
  - **Unit** feature was One-hot-encoded, missing **value** features were **filled with median** and **default IPQ value was taken as 1**.
  - A **`clean_text` feature** was created by **concatenating `item_name` and `bullet_points`** to serve as a noise-free input for the text encoder.
  - The text was tokenized using the **model-specific `AutoTokenizer`, with padding and truncation applied.**
- **Key parameters:**
  - **`max_length`**: 128 tokens

Shared Image Processing Pipeline:

- Images were mapped to corresponding sample_id using the **unique alphanumeric code** at the end of each image download link.
- **Missing or corrupt images** were handled by substituting a **blank white placeholder image.**
- Each **model's specific processor** (*`CLIPProcessor` or `AutoImageProcessor`*) was used to apply the required transformations (resize, center crop, and normalization) to the images.

Model 1: DistilBERT + ViT

- **Text Model:** `distilbert-base-uncased` (~66 Million parameters)

- **Image Model:** `google/vit-base-patch16-224-in21k` (~86 Million parameters)
- **MLP regressor : trainable parameters(~`152 Million`)**
- **Total Approximate Parameters:** ~152 Million

Model 2: RoBERTa-Large + CLIP-Large

- **Text Model:** `roberta-large` (~355 Million parameters)
- **Image Model:** `openai/clip-vit-large-patch14` (Vision Transformer only, ~427 Million parameters)
- **MLP regressor : trainable parameters(~`782 Million`)**
- **Total Approximate Parameters:** ~782 Million

Model 3: DeBERTa-Large + CLIP-Large

- **Text Model:** `microsoft/deberta-v3-large` (~435 Million parameters)
- **Image Model:** `openai/clip-vit-large-patch14` (Vision Transformer only, ~427 Million parameters)
- **MLP regressor : trainable parameters(~`738 Million`)**
- **Total Approximate Parameters:** ~738 Million

Final Ensemble Model:

Using the following weights
- **Model 1 (submission_48.39_model.csv)**: 0.15
- **Model 2 (submission_45.55_model.csv)**: 0.20
- **Model 3 (submission_deberta_large_clip-large_44.06.csv):** 0.65

We took Weighted Average of (Model 1, Model 2, Model 3)

4. Model Performance

4.1 Validation Results

Final Ensemble SMAPE Score: 43.5961%
Individual Model Validation SMAPE Scores:

- Model 1 (DistilBERT + ViT): **48.23%**
- Model 2 (RoBERTa-Large + CLIP-Large): **45.55%**
- Model 3 (DeBERTa-Large + CLIP-Large): **44.06%**

5. Conclusion

We systematically progressed from a single-model baseline to a final, optimized ensemble of three powerful multi-modal architectures, culminating in a best validation SMAPE of 43.5961%. Our key success was in the final ensemble step, which confirmed that combining predictions from diverse models like DeBERTa-Large and RoBERTa-Large provides a significant performance lift. The main lesson learned is that for complex prediction tasks, a carefully weighted ensemble is a critical tool for maximizing accuracy.

Appendix
Drive Link of complete code directory :
[Drive link]